

ChE 381: Unit Operations Laboratory

EVAPORATION

rev: Fall 2014

Introduction

The objective of evaporation is to concentrate a solution consisting of a nonvolatile solute and a volatile solvent. In the overwhelming majority of evaporations the solvent is water. Evaporation is conducted by vaporizing a portion of the solvent to produce a concentrated solution or thick liquor. Therefore, this is basically a heat transfer unit operation.

Evaporation differs from drying in that the residue is a liquid (sometimes a highly viscous one) rather than a solid. It differs from distillation in that the vapor is usually a single component, and even when the vapor is a mixture no attempt is made in the evaporation step to separate the vapor into fractions. It differs from crystallization in that emphasis is placed on concentrating a solution rather than forming and building crystals. In certain situations, e.g., in the evaporation of brine to produce common salt, the line between evaporation and crystallization is far from sharp. Evaporation sometimes produces a slurry of crystals in a saturated mother liquor.

Normally, in evaporation the thick liquor is the valuable product and the vapor is condensed and discarded. In one specific situation, however, the reverse is true. Mineral-bearing water often is evaporated to give a solid-free product for boiler feed, for special process requirements, or for human consumption. This process is often called water distillation, but technically it is evaporation.

Liquid Characteristics

The practical solution of an evaporation problem is profoundly affected by the character of the liquor to be concentrated. It is the wide variation in liquor characteristics (which demands judgment and experience in engineering and operating evaporators) that broadens this operation from the simple heat transfer to the separate art. Some of the most important properties of evaporating liquids are as follows.

Concentration

Although the thin liquor fed to an evaporator may be sufficiently dilute to have many of the physical properties of water, as the concentration increases, the solution becomes more and more distinct from water. The density and viscosity increase with the solid content, until either the solution becomes saturated or the liquor becomes too sluggish for adequate heat transfer. Continued boiling of a saturated solution causes crystals to form which must be removed to prevent clogging. The boiling point of the solution may also rise considerably as the solid content increases which may significantly decrease the heat transfer rate.

Foaming

Some materials, especially organic substances, foam during vaporization. A stable foam accompanies the vapor out of the evaporator, causing heavy entrainment. In extreme cases the entire mass of liquid may boil over into the vapor outlet and be lost.

Temperature sensitivity

Many fine chemicals, pharmaceutical products, and foods are damaged when heated to moderate temperatures for relatively short times. In concentrating such materials special techniques are needed to reduce both the temperature of the liquid and the time of heating.

Scale

Some solutions deposit scale on the heating surfaces. The over-all coefficient then steadily diminishes, until the evaporator must be shut down and the tubes cleaned. When the scale is hard and insoluble, the cleaning is difficult and expensive.

Materials of construction

Whenever possible, evaporators are made of cast iron and steel. Many solutions, however, attack ferrous metals or are contaminated by them. Special materials such as copper, nickel, stainless steel, aluminum, impervious graphite, and lead are then used. Since these materials are expensive high heat-transfer rates become especially desirable to minimize the capital cost of the equipment.

Many other liquid characteristics must be considered by the designer of an evaporator. Some of these are specific heat, heat of concentration, freezing point, gas liberation on boiling, toxicity, explosion hazards, radioactivity, and necessity for sterile operation. Because of the variation in liquor properties, many different evaporator designs have been developed. The choice for any specific problem depends primarily on the characteristics of the liquid.

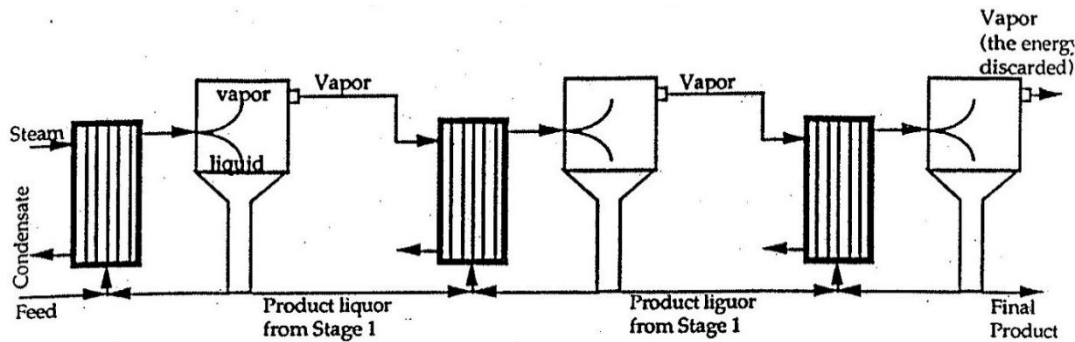
Single-and multiple-effect operation

Most evaporators are heated by steam which condenses on metal tubes. Most often the material to be evaporated flows inside the tubes. Usually the steam is at a low pressure, below 2000 mmHg (40 psia); and often the boiling liquid is under a moderate vacuum, up to about 700 mmHg of vacuum. The vacuum reduces the boiling temperature of the liquid, and increases the temperature difference between the steam and the boiling liquid. This increases the heat-transfer rate in the evaporator.

When a single evaporator is used, the vapor from the boiling liquid is condensed and discarded. This method is called single-effect evaporation, and although it is simple, it utilizes steam ineffectively. About 1 to 1.3 kg of steam is required for each 1 kg of water which evaporates. However, the vapor from the evaporator still contains energy. Therefore, the vapor from one evaporator can be fed into the steam chest of a second evaporator. This vapor can then be sent to a third evaporator, etc. until the vapor has lost a significant amount of energy. This is referred to as multiple-effect evaporation.

The heat in the original steam is reused in the subsequent effects, and the evaporation achieved by the original 1 kg of steam is theoretically multiplied by the number of effects. (See *Figure 1*) Notice that the feed to one stage is the product from the previous stage. Also, each effect must operate at a lower pressure to provide for a sufficient driving force for the heat transfer.

Figure 1: Multiple Effect Evaporation



The principle measures of the performance of a steam-heated tubular evaporator are the **capacity** and the **economy**. Capacity is defined as the number of kilograms of water vaporized of steam fed to the unit. In a single-effect evaporator the economy is nearly always less than 1, but in multiple-effect equipment it may be considerably greater. The steam consumption in kg per hour, is also important. It equals the capacity divided by the economy.

Design Equations

The evaporator is basically a heat transfer operation. In order to calculate the capacity and economy, we merely must perform an energy balance on the evaporator. The following discussion is for a single effect evaporator.

The rate of heat transfer to an evaporator is given by:

$$q = UA\Delta T \quad (1)$$

Where

- U = the overall heat transfer coefficient
- A = the area available for the heat transfer
- ΔT = the average temperature difference for heat transfer (this is normally uniform over the length of the evaporator since one side is condensing steam and the other side is boiling fluid)

If the feed to the evaporator is at the boiling temperature (at the given pressure), all the heat transferred through the heating surface is available for evaporation, and the capacity is proportional to " q ". If the feed is cold, the heat required to heat it to its boiling point may be quite large, and the capacity for a given value of " q " is reduced accordingly, as heat used to heat the feed is not available for evaporation. Conversely, if the feed is at a temperature above the boiling point in the vapor space, a portion of the

feed evaporates spontaneously by adiabatic equilibration with the vapor-space pressure, and the capacity is greater than that corresponding to “ q ”. This process is called flash evaporation. Therefore, “ q ” should be corrected for the feed temperature and converted to the desired units of kg of steam per hour by using the heat of vaporization of the steam at the operating pressure.

The actual temperature drop across the heating surface depends on the solution being evaporated, the difference in pressure between the steam and the boiling liquid, and the depth of liquid over the heating surface. In some evaporators the velocity of the liquid in the tubes also influences the temperature drop, because the frictional loss in the tubes increases the effective pressure of the liquid. When the solution has the characteristics of pure water, its boiling point can be read from the steam tables if the pressure is known, as can the temperature of the condensing steam. In actual evaporators, however the boiling point of a solution is affected by two factors, boiling-point elevation and liquid head.

The economy of an evaporator was defined above. In theory, the economy should equal “1” (unity) if there are no energy losses in the evaporator, and if the feed enters at its boiling point.

Experimental Procedure

A schematic of the single effect evaporator used in this experiment is shown below. A vacuum pump is connected to the loop containing the solution; this controls the pressure (and hence the boiling point) in the separator. Also, thermocouples are connected at various points. The vapor is sent to a condenser which is not shown in the schematic below.

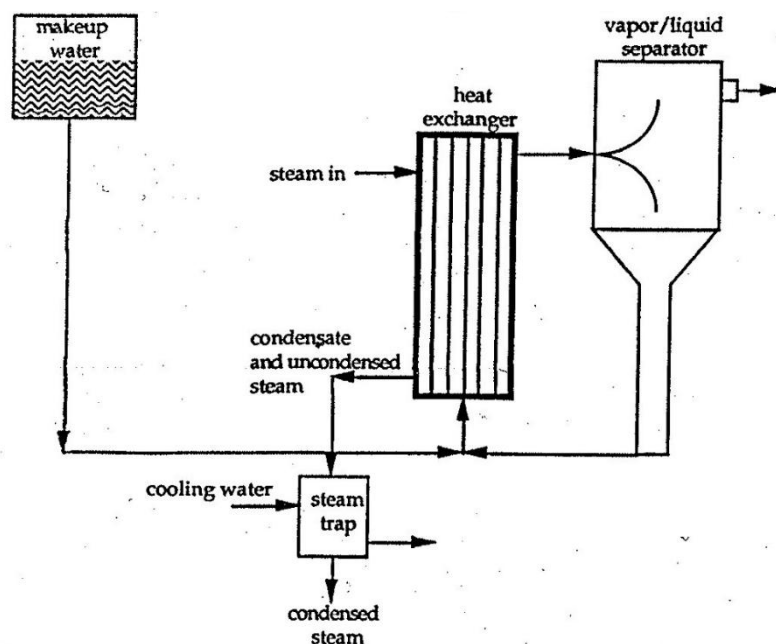


Figure 2: Schematic of the Single Effect Evaporator

- Drain contents of the separator through a valve at the bottom of the separator.
- Prepare a 10 wt% sugar solution, and fill the separator to the level line. Allow some of the solution to flow into the tube used to drain the separator.
- Put pure water in the makeup tank. During the experiment, you must adjust the flow of the makeup water so that the system stays at a constant level.
- Start the flow of steam.
- Start the cooling water to the steam trap. This will condense any remaining steam. Collect the steam condensate in a graduated cylinder.
- Turn on the vacuum pump, and adjust the bypass valve to obtain a desired pressure in the separator (the pressure gauge reads in inches of mercury). Notice that the solution will start circulating through the heat exchanger since a temperature difference now exists for the exchange of heat. Adjust the vacuum so that a steady flow is achieved (but not too high). You will perform the experiment at several vacuum pressure settings as discussed below.
- Collect the vapor product (after it has condensed). **Be sure to add enough makeup water to maintain the level at a steady state.** Notice that we are feeding pure water to this system. This is a little strange since we are removing pure water (in the vapor) and adding pure water (in the makeup). Therefore, the concentration of the solution is not changing. An industrial evaporator would feed a dilute solution, and continuously remove part of the concentrated solution.
- Wait for the system to reach a steady state as indicated by the temperature readings.
- After reaching a steady state, note the level of the steam condensate and collected vapor. Now, wait for 15 minutes (or so) and read these levels again. Use this to calculate the flowrates of these two streams (i.e., measure the time required to collect a given volume of fluid).
- Record the flow rate of the cooling water to the steam trap, and the inlet and outlet temperature. You need this in order to determine how much of the steam condensed in the heat exchanger (which then provided this energy to the solution).
- Now, change the pump to a different vacuum level, for a total of three vacuum settings. This will change the boiling point of the solution (why) and hence should increase the temperature difference between the solution and the steam. This should increase the capacity (but what about the economy?).

Data Analysis

Perform an energy balance on the steam and the solution in the evaporator. (I suggest that you draw the appropriate control volumes on some scratch paper in order to do this correctly. Chemical engineers must always know what control volume they are using.)

First, calculate the **capacity** by calculating the amount of product (condensate) collected per time (*kg/hour*) at the different vacuum pressures.

Now, calculate the **economy** as the ratio of (*kg of product/kg of steam used*) at the different vacuum pressures.

Explain any trends you observe in the performance of the evaporator when varying the vacuum setting. For example, what is the temperature difference between the steam and the solution at each vacuum setting? Does the capacity correlate with the temperature difference? How would you use the economy? Also, how does the boiling point of the sugar solution change with vacuum pressure?

Thought Questions

1. Does the heat of vaporization of steam depend on pressure? Where is the heat of vaporization found on a steam table?
2. What other factors besides the vacuum setting would affect the economy or the capacity of the evaporator in this lab? Sugar concentration? Steam flowrate? Liquid level in the separator?

References

1. McCabe W.L., Smith J.C., Harriott P. , "Unit Operations of Chemical Engineering", Chapter 16, 7th Edition, McGraw-Hill, NY, NY, 2005.